

NKPC HSA Bayesian State-Space Estimation

June 22, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is $N_t = 100 \log(N_t^{level})$ unless a run metadata file explicitly states that the series was already transformed.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + \kappa x_t - \theta\hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t - (\theta_0 + \gamma\bar{N}_t)\hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1\hat{N}_{t-1} + \rho_2\hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ , κ_0 , and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA_SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata.

4 Prior Specifications

Baseline, weak, and tight priors are stored in `configs/`. Robustness tables below summarize whether conclusions for δ , θ , and γ change across these choices.

5 1. Posterior Mean Coefficients

This section reports posterior means and credible intervals for the main scalar coefficients in each estimated run.

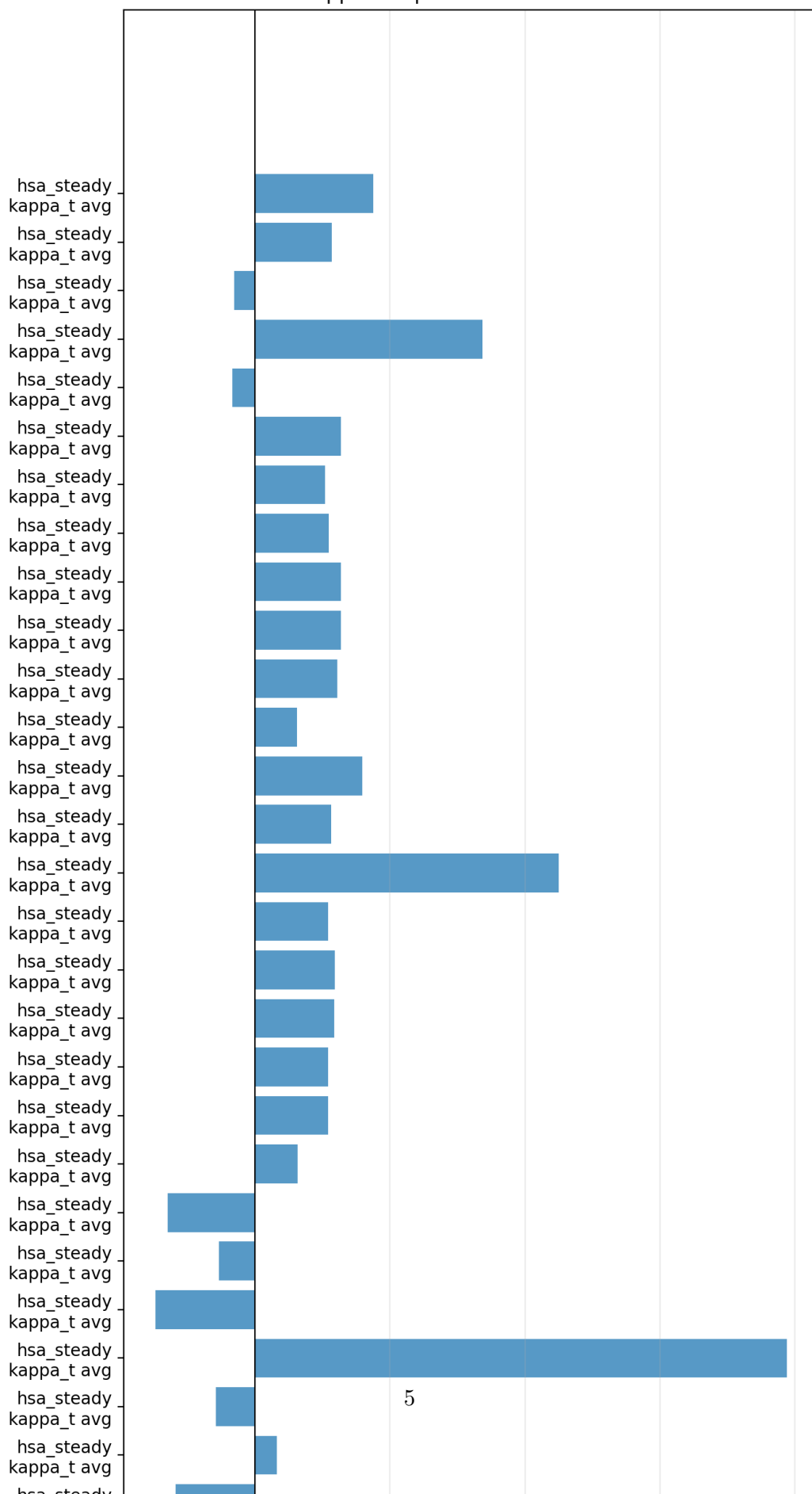
run	model	parameter	posterior_mean	posterior_sd	ci_2.5	ci_97.5
ces_default_baseline.20260622.011831	ces	alpha	0.6692	0.0692	0.5437	0.7835
ces_default_baseline.20260622.011831	ces	kappa	0.0623	0.0493	-0.0059	0.1764
ces_default_baseline.20260622.011831	ces	phi.1	0.9192	0.0369	0.8378	0.9703
ces_default_baseline.20260622.011831	ces	lambda.ez	-0.0085	0.1099	-0.2047	0.2214
ces_default_baseline.20260622.011832	ces	alpha	0.6692	0.0692	0.5437	0.7835
ces_default_baseline.20260622.011832	ces	kappa	0.0623	0.0493	-0.0059	0.1764
ces_default_baseline.20260622.011832	ces	phi.1	0.9192	0.0369	0.8378	0.9703
ces_default_baseline.20260622.011832	ces	lambda.ez	-0.0085	0.1099	-0.2047	0.2214
ces_default_baseline.20260622.011851	ces	alpha	0.6618	0.0674	0.5262	0.7745
ces_default_baseline.20260622.011851	ces	kappa	0.0472	0.0463	-0.0376	0.1371
ces_default_baseline.20260622.011851	ces	phi.1	0.9233	0.0270	0.8800	0.9713
ces_default_baseline.20260622.011851	ces	lambda.ez	-0.0003	0.1138	-0.1837	0.2104
ces_default_baseline.20260622.125732	ces	alpha	0.6692	0.0692	0.5437	0.7835
ces_default_baseline.20260622.125732	ces	kappa	0.0623	0.0493	-0.0059	0.1764
ces_default_baseline.20260622.125732	ces	phi.1	0.9192	0.0369	0.8378	0.9703
ces_default_baseline.20260622.125732	ces	lambda.ez	-0.0085	0.1099	-0.2047	0.2214
ces_default_baseline.20260622.130150	ces	alpha	0.6692	0.0692	0.5437	0.7835
ces_default_baseline.20260622.130150	ces	kappa	0.0623	0.0493	-0.0059	0.1764
ces_default_baseline.20260622.130150	ces	phi.1	0.9192	0.0369	0.8378	0.9703
ces_default_baseline.20260622.130150	ces	lambda.ez	-0.0085	0.1099	-0.2047	0.2214
ces_default_tight.20260622.011851	ces	alpha	0.6389	0.0506	0.5556	0.7145
ces_default_tight.20260622.011851	ces	kappa	0.0622	0.0472	-0.0206	0.1330
ces_default_tight.20260622.011851	ces	phi.1	0.8901	0.0280	0.8405	0.9390
ces_default_tight.20260622.011851	ces	lambda.ez	-0.0173	0.0917	-0.1794	0.1755
ces_default_weak.20260622.011851	ces	alpha	0.6802	0.0691	0.5556	0.7659
ces_default_weak.20260622.011851	ces	kappa	0.0581	0.0481	-0.0212	0.1583
ces_default_weak.20260622.011851	ces	phi.1	0.9198	0.0300	0.8687	0.9738
ces_default_weak.20260622.011851	ces	lambda.ez	0.0025	0.1087	-0.1814	0.1744
ces_inv_markup_baseline.20260622.131545	ces	alpha	0.6342	0.0588	0.5474	0.7428
ces_inv_markup_baseline.20260622.131545	ces	kappa	0.0116	0.1238	-0.2051	0.3018
ces_inv_markup_baseline.20260622.131545	ces	phi.1	0.6733	0.0721	0.5321	0.7739
ces_inv_markup_baseline.20260622.131545	ces	lambda.ez	-0.0370	0.1484	-0.3385	0.2248
ces_output_gap_bn_baseline.20260622.131732	ces	alpha	0.6692	0.0692	0.5437	0.7835
ces_output_gap_bn_baseline.20260622.131732	ces	kappa	0.0623	0.0493	-0.0059	0.1764
ces_output_gap_bn_baseline.20260622.131732	ces	phi.1	0.9192	0.0369	0.8378	0.9703
ces_output_gap_bn_baseline.20260622.131732	ces	lambda.ez	-0.0085	0.1099	-0.2047	0.2214
ces_unemployment_gap_baseline.20260622.131739	ces	alpha	0.6427	0.0616	0.5494	0.7623
ces_unemployment_gap_baseline.20260622.131739	ces	kappa	0.0577	0.0369	0.0063	0.1398
ces_unemployment_gap_baseline.20260622.131739	ces	phi.1	0.9533	0.0306	0.8906	0.9957
ces_unemployment_gap_baseline.20260622.131739	ces	lambda.ez	0.0276	0.0971	-0.1368	0.1816
hsa_dynamic_default_baseline.20260622.125743	hsa_dynamic	alpha	0.6417	0.0572	0.5407	0.7302
hsa_dynamic_default_baseline.20260622.125743	hsa_dynamic	kappa	0.0812	0.0406	-0.0037	0.1503
hsa_dynamic_default_baseline.20260622.125743	hsa_dynamic	theta	0.0089	0.0035	0.0011	0.0144
hsa_dynamic_default_baseline.20260622.125743	hsa_dynamic	phi.1	0.9069	0.0263	0.8613	0.9529
hsa_dynamic_default_baseline.20260622.125743	hsa_dynamic	rho.1	1.8622	0.0360	1.7937	1.9130
hsa_dynamic_default_baseline.20260622.125743	hsa_dynamic	rho.2	-0.8657	0.0358	-0.9167	-0.7966
hsa_dynamic_default_baseline.20260622.125743	hsa_dynamic	n	-0.0263	0.0733	-0.1502	0.1060
hsa_dynamic_default_baseline.20260622.130153	hsa_dynamic	alpha	0.6417	0.0572	0.5407	0.7302
hsa_dynamic_default_baseline.20260622.130153	hsa_dynamic	kappa	0.0812	0.0406	-0.0037	0.1503
hsa_dynamic_default_baseline.20260622.130153	hsa_dynamic	theta	0.0089	0.0035	0.0011	0.0144
hsa_dynamic_default_baseline.20260622.130153	hsa_dynamic	phi.1	0.9069	0.0263	0.8613	0.9529
hsa_dynamic_default_baseline.20260622.130153	hsa_dynamic	rho.1	1.8622	0.0360	1.7937	1.9130
hsa_dynamic_default_baseline.20260622.130153	hsa_dynamic	rho.2	-0.8657	0.0358	-0.9167	-0.7966
hsa_dynamic_default_baseline.20260622.130153	hsa_dynamic	n	-0.0263	0.0733	-0.1502	0.1060
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	alpha	0.6161	0.0515	0.5294	0.7235
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	kappa	0.0301	0.1531	-0.2127	0.3024
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	theta	0.0083	0.0039	0.0029	0.0158
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	phi.1	0.6507	0.0710	0.5172	0.7831
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	rho.1	1.8713	0.0303	1.8074	1.9099
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	rho.2	-0.8748	0.0301	-0.9150	-0.8101
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	n	-0.0353	0.0486	-0.1143	0.0509
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	alpha	0.6417	0.0572	0.5407	0.7302
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	kappa	0.0812	0.0406	-0.0037	0.1503
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	theta	0.0089	0.0035	0.0011	0.0144
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	phi.1	0.9069	0.0263	0.8613	0.9529
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	rho.1	1.8622	0.0360	1.7937	1.9130
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	rho.2	-0.8657	0.0358	-0.9167	-0.7966
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	n	-0.0263	0.0733	-0.1502	0.1060
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	alpha	0.5917	0.0454	0.5213	0.6629
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	kappa	0.0943	0.0348	0.0443	0.1599
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	theta	0.0117	0.0033	0.0063	0.0185
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	phi.1	0.9545	0.0264	0.9117	0.9997
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	rho.1	1.8723	0.0418	1.7904	1.9412
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	rho.2	-0.8759	0.0417	-0.9454	-0.7923
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	n	0.0007	0.0562	-0.0913	0.0972
hsa_full_default_baseline.20260622.125744	hsa_full	alpha	0.6624	0.0547	0.5625	0.7575
hsa_full_default_baseline.20260622.125744	hsa_full	kappa.0	0.2476	0.1843	-0.0392	0.5474
hsa_full_default_baseline.20260622.125744	hsa_full	delta	-0.0010	0.0010	-0.0023	0.0005
hsa_full_default_baseline.20260622.125744	hsa_full	theta.0	0.1009	0.1795	-0.1749	0.4086
hsa_full_default_baseline.20260622.125744	hsa_full	gamma	-0.0003	0.0009	-0.0017	0.0012
hsa_full_default_baseline.20260622.125744	hsa_full	phi.1	0.9163	0.0287	0.8724	0.9763
hsa_full_default_baseline.20260622.125744	hsa_full	rho.1	1.1912	0.1128	1.0032	1.3882
hsa_full_default_baseline.20260622.125744	hsa_full	rho.2	-0.2689	0.1196	-0.4702	-0.0667
hsa_full_default_baseline.20260622.125744	hsa_full	n	-0.0899	0.0590	-0.1806	0.0369
hsa_full_default_baseline.20260622.125744	hsa_full	lambda.ez	-0.0133	0.1240	-0.1946	0.2723
hsa_full_default_baseline.20260622.130155	hsa_full	alpha	0.6624	0.0547	0.5625	0.7575
hsa_full_default_baseline.20260622.130155	hsa_full	kappa.0	0.2476	0.1843	-0.0392	0.5474
hsa_full_default_baseline.20260622.130155	hsa_full	delta	-0.0010	0.0010	-0.0023	0.0005
hsa_full_default_baseline.20260622.130155	hsa_full	theta.0	0.1009	0.1795	-0.1749	0.4086
hsa_full_default_baseline.20260622.130155	hsa_full	gamma	-0.0003	0.0009	-0.0017	0.0012
hsa_full_default_baseline.20260622.130155	hsa_full	phi.1	0.9163	0.0287	0.8724	0.9763
hsa_full_default_baseline.20260622.130155	hsa_full	rho.1	1.1912	0.1128	1.0032	1.3882
hsa_full_default_baseline.20260622.130155	hsa_full	rho.2	-0.2689	0.1196	-0.4702	-0.0667

6 2. Kappa Comparison Across Models

For CES models this is κ . For steady or full HSA models the table also reports κ_0 , δ , and the average path of $\kappa_t = \kappa_0 + \delta \bar{N}_t$ when available.

run	model	kappa_0_2000	kappa_0_2.5	kappa_0_5	kappa_0_1000	kappa_0_2.5	kappa_0_5_97.5	delta_0_2000	delta_0_2.5	delta_0_5	kappa_0_0verall_2000	kappa_0_start_2000	kappa_0_end_2000
ces.default_baseline_20200622,111831	ces	0.0623	-0.0059	0.1764	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ces.default_baseline_20200622,111832	ces	0.0623	-0.0059	0.1764	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ces.default_baseline_20200622,111851	ces	0.0472	-0.0376	0.1371	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ces.default_baseline_20200622,125732	ces	0.0623	-0.0059	0.1764	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ces.default_baseline_20200622,130150	ces	0.0623	-0.0059	0.1764	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ces.default_tight_20200622,111851	ces	0.0622	-0.0206	0.1330	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ces.default_weak_20200622,111851	ces	0.0581	-0.0212	0.1583	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ces.inv.markup_baseline_20200622,131545	ces	0.0116	-0.2051	0.3018	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ces.output_gap_hsa_baseline_20200622,131732	ces	0.0623	-0.0059	0.1764	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ces.unemployment_gap_baseline_20200622,131739	ces	0.0577	0.0063	0.1398	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.default_baseline_20200622,125743	hsa_dynamic	0.0812	-0.0037	0.1503	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.default_baseline_20200622,130153	hsa_dynamic	0.0812	-0.0037	0.1503	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.inv.markup_baseline_20200622,131549	hsa_dynamic	0.0301	-0.2127	0.3024	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.output_gap_hsa_baseline_20200622,131735	hsa_dynamic	0.0812	-0.0037	0.1503	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa.dynamic.unemployment_gap_baseline_20200622,131740	hsa_dynamic	0.0943	0.0443	0.1599	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa.full.default_baseline_20200622,125744	hsa_full	NaN	NaN	0.2476	-0.0392	0.5474	-0.0010	-0.0023	0.0005	0.0485	0.0531	0.0745	0.0745
hsa.full.default_baseline_20200622,130155	hsa_full	NaN	NaN	0.2476	-0.0392	0.5474	-0.0010	-0.0023	0.0005	0.0485	0.0531	0.0745	0.0745
hsa.full.inv.markup_baseline_20200622,131551	hsa_full	NaN	NaN	0.0961	-0.2090	0.4047	-0.0004	-0.0028	0.0013	0.0409	0.0606	0.0163	0.0163
hsa.full.output_gap_hsa_baseline_20200622,131737	hsa_full	NaN	NaN	0.2476	-0.0392	0.5474	-0.0010	-0.0023	0.0005	0.0485	0.0531	0.0745	0.0745
hsa.full.unemployment_gap_hsa_baseline_20200622,131742	hsa_full	NaN	NaN	-0.0501	-0.3008	0.2173	0.0006	-0.0008	0.0026	0.0644	0.0620	0.0485	0.0485
hsa.steady.default_baseline_20200622,125741	hsa_steady	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567	0.0567
hsa.steady.default_baseline_20200622,125752	hsa_steady	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567	0.0567
hsa.steady.default_baseline_20200622,125759	hsa_steady	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567	0.0567
hsa.steady.default_baseline_20200622,125802	hsa_steady	NaN	NaN	0.1203	-0.1578	0.4917	-0.0003	-0.0023	0.0013	0.0563	0.0563	0.0565	0.0565
hsa.steady.default_baseline_20200622,125804	hsa_steady	NaN	NaN	0.1573	-0.1098	0.4860	-0.0005	-0.0020	0.0009	0.0591	0.0552	0.0631	0.0631
hsa.steady.default_baseline_20200622,125806	hsa_steady	NaN	NaN	0.0971	-0.2810	0.4612	-0.0001	-0.0021	0.0017	0.0794	0.0772	0.0823	0.0823
hsa.steady.default_baseline_20200622,125808	hsa_steady	NaN	NaN	0.1099	-0.1449	0.4737	-0.0003	-0.0022	0.0012	0.0588	0.0590	0.0587	0.0587
hsa.steady.default_baseline_20200622,130151	hsa_steady	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567	0.0567
hsa.steady.default_baseline_20200622,130219	hsa_steady	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567	0.0567
hsa.steady.default_baseline_20200622,130228	hsa_steady	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567	0.0567
hsa.steady.default_baseline_20200622,130231	hsa_steady	NaN	NaN	0.1203	-0.1578	0.4917	-0.0003	-0.0023	0.0013	0.0563	0.0563	0.0565	0.0565
hsa.steady.default_baseline_20200622,130233	hsa_steady	NaN	NaN	0.1573	-0.1098	0.4860	-0.0005	-0.0020	0.0009	0.0591	0.0552	0.0631	0.0631
hsa.steady.default_baseline_20200622,130235	hsa_steady	NaN	NaN	0.0971	-0.2810	0.4612	-0.0001	-0.0021	0.0017	0.0794	0.0772	0.0823	0.0823
hsa.steady.default_baseline_20200622,130236	hsa_steady	NaN	NaN	0.1099	-0.1449	0.4737	-0.0003	-0.0022	0.0012	0.0588	0.0590	0.0587	0.0587
hsa.steady.default_tight_20200622,125756	hsa_steady	NaN	NaN	0.1386	-0.0929	0.3230	-0.0005	-0.0020	0.0009	0.0310	0.0322	0.0451	0.0451
hsa.steady.default_tight_20200622,130224	hsa_steady	NaN	NaN	0.1386	-0.0929	0.3230	-0.0005	-0.0020	0.0009	0.0310	0.0322	0.0451	0.0451
hsa.steady.default_weak_20200622,125755	hsa_steady	NaN	NaN	0.3115	-0.3883	1.0701	-0.0013	-0.0054	0.0028	0.0608	0.0449	0.0770	0.0770
hsa.steady.default_weak_20200622,130222	hsa_steady	NaN	NaN	0.3115	-0.3883	1.0701	-0.0013	-0.0054	0.0028	0.0608	0.0449	0.0770	0.0770
hsa.steady.inv.markup_baseline_20200622,131547	hsa_steady	NaN	NaN	0.1008	-0.1878	0.4623	-0.0006	-0.0028	0.0021	-0.0290	-0.0294	-0.0286	-0.0286
hsa.steady.inv.markup_baseline_20200622,132032	hsa_steady	NaN	NaN	0.1008	-0.1878	0.4623	-0.0006	-0.0028	0.0021	-0.0290	-0.0294	-0.0286	-0.0286
hsa.steady.inv.markup.post.2008_baseline_20200622,132122	hsa_steady	NaN	NaN	0.1276	-0.2432	0.5406	-0.0009	-0.0036	0.0016	-0.0587	-0.0612	-0.0544	-0.0544
hsa.steady.inv.markup.post.covid_baseline_20200622,132118	hsa_steady	NaN	NaN	0.0867	-0.2233	0.5577	-0.0003	-0.0030	0.0017	0.0163	0.0170	0.0159	0.0159
hsa.steady.inv.markup.full_baseline_20200622,132113	hsa_steady	NaN	NaN	0.1008	-0.1878	0.4623	-0.0006	-0.0028	0.0021	-0.0290	-0.0294	-0.0286	-0.0286
hsa.steady.inv.markup.post.2008_baseline_20200622,132118	hsa_steady	NaN	NaN	0.1268	-0.2747	0.6143	0.0015	-0.0036	0.0005	0.3942	0.3972	0.3914	0.3914
hsa.steady.inv.markup.post.2008_baseline_20200622,132117	hsa_steady	NaN	NaN	0.1268	-0.2639	0.5785	-0.0010	-0.0031	0.0009	-0.0788	-0.0731	-0.0738	-0.0738
hsa.steady.inv.markup.start.1988_baseline_20200622,132120	hsa_steady	NaN	NaN	0.0692	-0.2709	0.4122	-0.0005	-0.0024	0.0016	-0.0270	-0.0310	-0.0280	-0.0280
hsa.steady.inv.markup.tight_20200622,132037	hsa_steady	NaN	NaN	0.0996	-0.1344	0.2868	-0.0008	-0.0023	0.0005	-0.0649	-0.0628	-0.0433	-0.0433
hsa.steady.inv.markup_weak_20200622,132035	hsa_steady	NaN	NaN	0.2963	-0.7736	1.3392	-0.0009	-0.0066	0.0041	0.0315	0.0217	0.0411	0.0411
hsa.steady.output_gap_hsa_baseline_20200622,131733	hsa_steady	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567	0.0567
hsa.steady.output_gap_hsa_baseline_20200622,132124	hsa_steady	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567	0.0567
hsa.steady.output_gap_hsa.post.2008_baseline_20200622,132131	hsa_steady	NaN	NaN	0.1099	-0.1449	0.4737	-0.0003	-0.0022	0.0012	0.0588	0.0590	0.0587	0.0587
hsa.steady.output_gap_hsa.start.covid_baseline_20200622,132128	hsa_steady	NaN	NaN	0.1573	-0.1098	0.4860	-0.0005	-0.0020	0.0009	0.0591	0.0552	0.0631	0.0631
hsa.steady.output_gap_hsa.full_baseline_20200622,132134	hsa_steady	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567	0.0567
hsa.steady.output_gap_hsa.post.2008_baseline_20200622,132127	hsa_steady	NaN	NaN	0.1335	-0.2710	0.6127	0.0005	-0.0024	0.0027	0.2252	0.2364	0.2242	0.2242
hsa.steady.output_gap_hsa.post.2008_baseline_20200622,132125	hsa_steady	NaN	NaN	0.1203	-0.1578	0.4917	-0.0003	-0.0023	0.0013	0.0563	0.0563	0.0565	0.0565
hsa.steady.output_gap_hsa.tight_20200622,132130	hsa_steady	NaN	NaN	0.0971	-0.2810	0.4612	-0.0001	-0.0021	0.0017	0.0794	0.0772	0.0823	0.0823
hsa.steady.output_gap_hsa.tight_20200622,132047	hsa_steady	NaN	NaN	0.1386	-0.0929	0.3230	-0.0005	-0.0020	0.0009	0.0310	0.0322	0.0451	0.0451
hsa.steady.output_gap_hsa_weak_20200622,132044	hsa_steady	NaN	NaN	0.3115	-0.3883	1.0701	-0.0013	-0.0054	0.0028	0.0608	0.0449	0.0770	0.0770
hsa.steady.unemployment_gap_baseline_20200622,131739	hsa_steady	NaN	NaN	0.0757	-0.3673	0.3665	-0.0001	-0.0017	0.0021	0.0636	0.0638	0.0629	0.0629
hsa.steady.unemployment_gap_baseline_20200622,132052	hsa_steady	NaN	NaN	0.0757	-0.3673	0.3665	-0.0001	-0.0017	0.0021	0.0636	0.0638	0.0629	0.0629
hsa.steady.unemployment_gap.post.2019_baseline_20200622,132140	hsa_steady	NaN	NaN	0.0042	-0.4906	0.3742	0.0003	-0.0016	0.0026	0.0547	0.0568	0.0526	0.0526
hsa.steady.unemployment_gap.start.covid_baseline_20200622,132137	hsa_steady	NaN	NaN	-0.0200	-0.4271	0.2862	0.0004	-0.0012	0.0026	0.0518	0.0561	0.0467	0.0467
hsa.steady.unemployment_gap.full_baseline_20200622,132133	hsa_steady	NaN	NaN	0.0757	-0.3673	0.3665	-0.0001	-0.0017	0.0021	0.0636	0.0638	0.0629	0.0629
hsa.steady.unemployment_gap.post.2008_baseline_20200622,132137	hsa_steady	NaN	NaN	0.0674	-0.4251	0.4638	-0.0005	-0.0030	0.0021	-0.0169	-0.0180	-0.0180	-0.0180
hsa.steady.unemployment_gap.post.2008_baseline_20200622,132135	hsa_steady	NaN	NaN	0.0984	-0.1924	0.4616	0.0003	-0.0016	0.0021	0.1686	0.1692	0.1679	0.1679
hsa.steady.unemployment_gap.start.1988_baseline_20200622,132139	hsa_steady	NaN	NaN	0.0526	-0.4687	0.4666	-0.0003	-0.0026	0.0023	-0.0156	-0.0203	-0.0106	-0.0106
hsa.steady.unemployment_gap.tight_20200622,132056	hsa_steady	NaN	NaN	0.0459	-0.1360	0.2705	0.0001	-0.0009	0.0009	0.0567	0.0564	0.0552	0.0552
hsa.steady.unemployment_gap_weak_20200622,132053	hsa_steady	NaN	NaN	-0.6887	-1.1713	-0.1801	0.0041	0.0015	0.0066	0.0876	0.1564	0.0165	0.0165

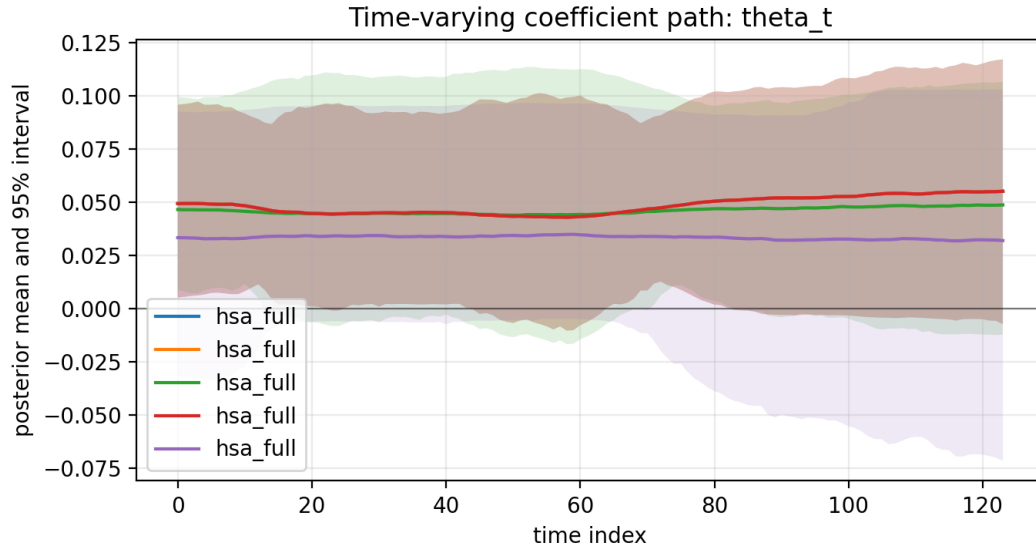
Kappa comparison across models



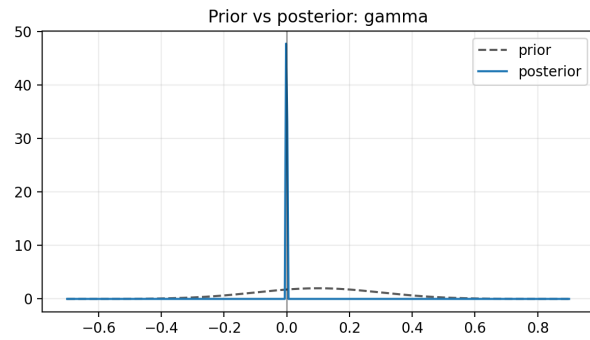
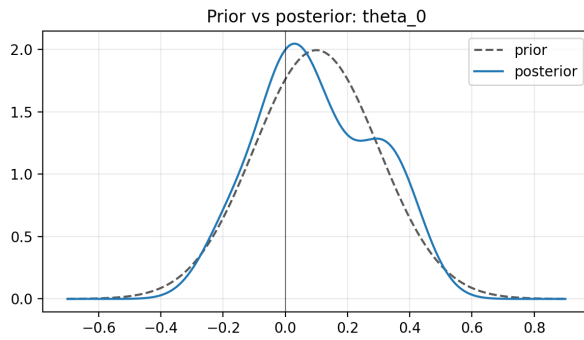
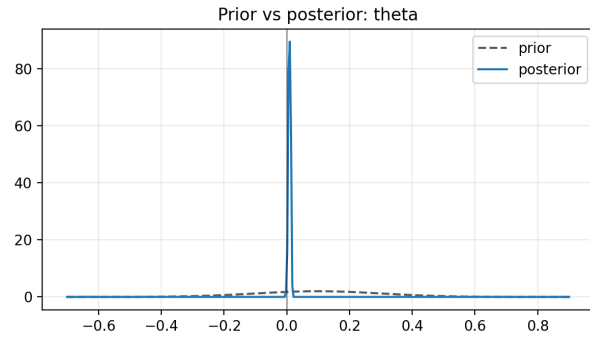
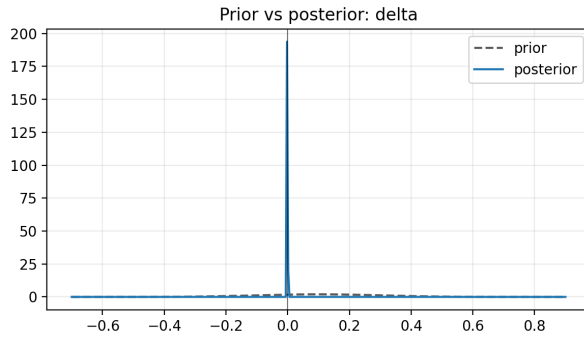
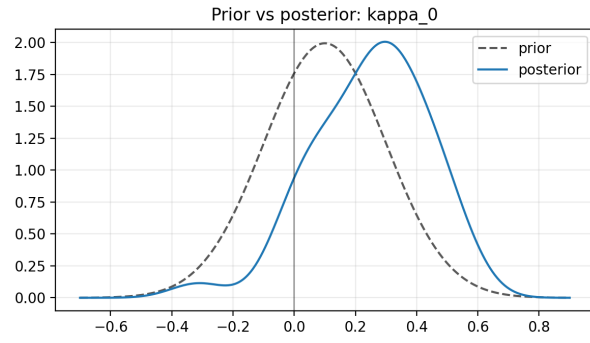
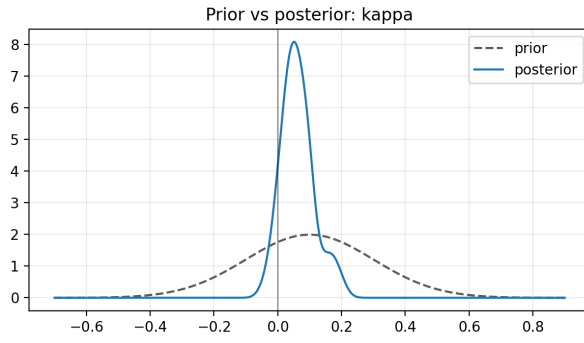
7 3. Time-Varying Coefficient Paths

The main time-varying coefficients are κ_t and, in the both-effects model, $\theta_t = \theta_0 + \gamma\bar{N}_t$.

run	model	coefficient	time_index	posterior_mean	ci_2.5	ci_97.5
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	0	0.0531	-0.0205	0.1427
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	5	0.0524	-0.0247	0.1445
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	11	0.0473	-0.0336	0.1347
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	16	0.0380	-0.0469	0.1201
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	21	0.0352	-0.0518	0.1265
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	27	0.0361	-0.0496	0.1224
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	32	0.0369	-0.0497	0.1236
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	37	0.0376	-0.0476	0.1223
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	43	0.0351	-0.0516	0.1262
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	48	0.0321	-0.0560	0.1233
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	53	0.0303	-0.0591	0.1204
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	59	0.0301	-0.0601	0.1179
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	64	0.0349	-0.0515	0.1246
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	70	0.0436	-0.0397	0.1272
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	75	0.0506	-0.0280	0.1383
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	80	0.0574	-0.0195	0.1478
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	86	0.0610	-0.0132	0.1544
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	91	0.0623	-0.0122	0.1588
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	96	0.0635	-0.0098	0.1591
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	102	0.0669	-0.0071	0.1642
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	107	0.0709	-0.0055	0.1695
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	112	0.0713	-0.0049	0.1696
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	118	0.0740	-0.0043	0.1732
hsa_full.default.baseline.20260622.125744	hsa_full	kappa.t	123	0.0745	-0.0040	0.1733
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	0	0.0493	0.0052	0.0959
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	5	0.0491	0.0073	0.0959
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	11	0.0478	0.0098	0.0903
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	16	0.0454	0.0019	0.0922
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	21	0.0446	0.0007	0.0957
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	27	0.0448	0.0015	0.0936
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	32	0.0450	0.0020	0.0923
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	37	0.0452	0.0019	0.0925
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	43	0.0447	0.0003	0.0948
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	48	0.0436	-0.0053	0.0975
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	53	0.0431	-0.0064	0.1008
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	59	0.0430	-0.0102	0.0996
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	64	0.0443	0.0004	0.0960
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	70	0.0468	0.0084	0.0887
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	75	0.0486	0.0088	0.0958
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	80	0.0505	0.0018	0.1021
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	86	0.0515	-0.0014	0.1041
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	91	0.0521	-0.0031	0.1040
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	96	0.0522	-0.0030	0.1047
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	102	0.0531	-0.0039	0.1092
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	107	0.0542	-0.0045	0.1137
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	112	0.0541	-0.0046	0.1130
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	118	0.0549	-0.0050	0.1151
hsa_full.default.baseline.20260622.125744	hsa_full	theta.t	123	0.0551	-0.0074	0.1172
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	0	0.0531	-0.0205	0.1427
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	5	0.0524	-0.0247	0.1445
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	11	0.0473	-0.0336	0.1347
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	16	0.0380	-0.0469	0.1201
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	21	0.0352	-0.0518	0.1265
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	27	0.0361	-0.0496	0.1224
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	32	0.0369	-0.0497	0.1236
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	37	0.0376	-0.0476	0.1223
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	43	0.0351	-0.0516	0.1262
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	48	0.0321	-0.0560	0.1233
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	53	0.0303	-0.0591	0.1204
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	59	0.0301	-0.0601	0.1179
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	64	0.0349	-0.0515	0.1246
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	70	0.0436	-0.0397	0.1272
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	75	0.0506	-0.0280	0.1383
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	80	0.0574	-0.0195	0.1478
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	86	0.0610	-0.0132	0.1544
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	91	0.0623	-0.0122	0.1588
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	96	0.0635	-0.0098	0.1591
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	102	0.0669	-0.0071	0.1642
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	107	0.0709	-0.0055	0.1695
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	112	0.0713	-0.0049	0.1696
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	118	0.0740	-0.0043	0.1732
hsa_full.default.baseline.20260622.130155	hsa_full	kappa.t	123	0.0745	-0.0040	0.1733
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	0	0.0493	0.0052	0.0959
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	5	0.0491	0.0073	0.0959
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	11	0.0478	0.0098	0.0903
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	16	0.0454	0.0019	0.0922
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	21	0.0446	0.0007	0.0957
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	27	0.0448	0.0015	0.0936
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	32	0.0450	0.0020	0.0923
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	37	0.0452	0.0019	0.0925
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	43	0.0447	0.0003	0.0948
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	48	0.0436	-0.0053	0.0975
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	53	0.0431	-0.0064	0.1008
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	59	0.0430	-0.0102	0.0996
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	64	0.0443	0.0004	0.0960
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	70	0.0468	0.0084	0.0887
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	75	0.0486	0.0088	0.0958
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	80	0.0505	0.0018	0.1021
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	86	0.0515	-0.0014	0.1041
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	91	0.0521	-0.0031	0.1040
hsa_full.default.baseline.20260622.130155	hsa_full	theta.t	96	0.0522	-0.0030	0.1047



run	model	restriction	prior_mean	prior_sd	sddr_bf01
ces_default_baseline.20260622.011831	ces	kappa=0	0.1000	0.2000	2.3727
ces_default_baseline.20260622.011832	ces	kappa=0	0.1000	0.2000	2.3727
ces_default_baseline.20260622.011851	ces	kappa=0	0.1000	0.2000	3.2331
ces_default_baseline.20260622.125732	ces	kappa=0	0.1000	0.2000	2.3727
ces_default_baseline.20260622.130150	ces	kappa=0	0.1000	0.2000	2.3727
ces_default_tight.20260622.011851	ces	kappa=0	0.1000	0.2000	2.1383
ces_default_weak.20260622.011851	ces	kappa=0	0.1000	0.2000	2.6782
ces_inv_markup_baseline.20260622.131545	ces	kappa=0	0.1000	0.2000	2.0791
ces_output_gap_bn_baseline.20260622.131732	ces	kappa=0	0.1000	0.2000	2.3727
ces_unemployment_gap_baseline.20260622.131739	ces	kappa=0	0.1000	0.2000	2.0389
hsa_dynamic.default_baseline.20260622.125743	hsa_dynamic	kappa=0	0.1000	0.2000	1.0740
hsa_dynamic.default_baseline.20260622.125743	hsa_dynamic	theta=0	0.1000	0.2000	6.4392
hsa_dynamic.default_baseline.20260622.130153	hsa_dynamic	kappa=0	0.1000	0.2000	1.0740
hsa_dynamic.default_baseline.20260622.130153	hsa_dynamic	theta=0	0.1000	0.2000	6.4392
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	kappa=0	0.1000	0.2000	1.3783
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	theta=0	0.1000	0.2000	4.5471
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	kappa=0	0.1000	0.2000	1.0740
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	theta=0	0.1000	0.2000	6.4392
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	kappa=0	0.1000	0.2000	0.1520
hsa_dynamic_unemployment_gap_baseline.20260622.131740	hsa_dynamic	theta=0	0.1000	0.2000	0.5418
hsa_full.default_baseline.20260622.125744	hsa_full	kappa_0=0	0.1000	0.2000	0.5349
hsa_full.default_baseline.20260622.125744	hsa_full	delta=0	0.1000	0.2000	126.7357
hsa_full.default_baseline.20260622.125744	hsa_full	theta_0=0	0.1000	0.2000	1.1348
hsa_full.default_baseline.20260622.125744	hsa_full	gamma=0	0.1000	0.2000	243.6010
hsa_full.default_baseline.20260622.130155	hsa_full	kappa_0=0	0.1000	0.2000	0.5349
hsa_full.default_baseline.20260622.130155	hsa_full	delta=0	0.1000	0.2000	126.7357
hsa_full.default_baseline.20260622.130155	hsa_full	theta_0=0	0.1000	0.2000	1.1348
hsa_full.default_baseline.20260622.130155	hsa_full	gamma=0	0.1000	0.2000	243.6010
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa_0=0	0.1000	0.2000	0.8905
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	delta=0	0.1000	0.2000	158.5492
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta_0=0	0.1000	0.2000	0.8677
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	gamma=0	0.1000	0.2000	180.6376
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa_0=0	0.1000	0.2000	0.5349
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	delta=0	0.1000	0.2000	126.7357
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta_0=0	0.1000	0.2000	1.1348
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	gamma=0	0.1000	0.2000	243.6010
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	kappa_0=0	0.1000	0.2000	1.0683
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	delta=0	0.1000	0.2000	187.3843
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	theta_0=0	0.1000	0.2000	0.9029
hsa_full_unemployment_gap_baseline.20260622.131742	hsa_full	gamma=0	0.1000	0.2000	204.9817
hsa_steady.default_baseline.20260622.125741	hsa_steady	kappa_0=0	0.1000	0.2000	0.6104
hsa_steady.default_baseline.20260622.125741	hsa_steady	delta=0	0.1000	0.2000	165.3272
hsa_steady.default_baseline.20260622.125752	hsa_steady	kappa_0=0	0.1000	0.2000	0.6104
hsa_steady.default_baseline.20260622.125752	hsa_steady	delta=0	0.1000	0.2000	165.3272
hsa_steady.default_baseline.20260622.125759	hsa_steady	kappa_0=0	0.1000	0.2000	0.6104
hsa_steady.default_baseline.20260622.125759	hsa_steady	delta=0	0.1000	0.2000	165.3272
hsa_steady.default_baseline.20260622.125802	hsa_steady	kappa_0=0	0.1000	0.2000	1.1270
hsa_steady.default_baseline.20260622.125802	hsa_steady	delta=0	0.1000	0.2000	256.7958
hsa_steady.default_baseline.20260622.125804	hsa_steady	kappa_0=0	0.1000	0.2000	0.9697
hsa_steady.default_baseline.20260622.125804	hsa_steady	delta=0	0.1000	0.2000	206.3119
hsa_steady.default_baseline.20260622.125806	hsa_steady	kappa_0=0	0.1000	0.2000	0.9310
hsa_steady.default_baseline.20260622.125806	hsa_steady	delta=0	0.1000	0.2000	210.5789
hsa_steady.default_baseline.20260622.125808	hsa_steady	kappa_0=0	0.1000	0.2000	1.0543
hsa_steady.default_baseline.20260622.125808	hsa_steady	delta=0	0.1000	0.2000	218.4506
hsa_steady.default_baseline.20260622.130151	hsa_steady	kappa_0=0	0.1000	0.2000	0.6104
hsa_steady.default_baseline.20260622.130151	hsa_steady	delta=0	0.1000	0.2000	165.3272
hsa_steady.default_baseline.20260622.130219	hsa_steady	kappa_0=0	0.1000	0.2000	0.6104
hsa_steady.default_baseline.20260622.130219	hsa_steady	delta=0	0.1000	0.2000	165.3272
hsa_steady.default_baseline.20260622.130228	hsa_steady	kappa_0=0	0.1000	0.2000	0.6104
hsa_steady.default_baseline.20260622.130228	hsa_steady	delta=0	0.1000	0.2000	165.3272
hsa_steady.default_baseline.20260622.130231	hsa_steady	kappa_0=0	0.1000	0.2000	1.1270
hsa_steady.default_baseline.20260622.130231	hsa_steady	delta=0	0.1000	0.2000	256.7958
hsa_steady.default_baseline.20260622.130233	hsa_steady	kappa_0=0	0.1000	0.2000	0.9697
hsa_steady.default_baseline.20260622.130233	hsa_steady	delta=0	0.1000	0.2000	206.3119
hsa_steady.default_baseline.20260622.130235	hsa_steady	kappa_0=0	0.1000	0.2000	0.9310
hsa_steady.default_baseline.20260622.130235	hsa_steady	delta=0	0.1000	0.2000	210.5789
hsa_steady.default_baseline.20260622.130236	hsa_steady	kappa_0=0	0.1000	0.2000	1.0543
hsa_steady.default_baseline.20260622.130236	hsa_steady	delta=0	0.1000	0.2000	218.4506
hsa_steady.default_tight.20260622.125756	hsa_steady	kappa_0=0	0.1000	0.2000	0.6489
hsa_steady.default_tight.20260622.125756	hsa_steady	delta=0	0.1000	0.2000	231.3719
hsa_steady.default_tight.20260622.130224	hsa_steady	kappa_0=0	0.1000	0.2000	0.6489
hsa_steady.default_tight.20260622.130224	hsa_steady	delta=0	0.1000	0.2000	231.3719
hsa_steady.default_weak.20260622.125755	hsa_steady	kappa_0=0	0.1000	0.2000	0.3411
hsa_steady.default_weak.20260622.125755	hsa_steady	delta=0	0.1000	0.2000	68.8500
hsa_steady.default_weak.20260622.130222	hsa_steady	kappa_0=0	0.1000	0.2000	0.3411
hsa_steady.default_weak.20260622.130222	hsa_steady	delta=0	0.1000	0.2000	68.8500
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa_0=0	0.1000	0.2000	1.1450
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	delta=0	0.1000	0.2000	144.3115
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	kappa_0=0	0.1000	0.2000	1.1450
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	delta=0	0.1000	0.2000	144.3115
hsa_steady_inv_markup_end.2019_baseline.20260622.132122	hsa_steady	kappa_0=0	0.1000	0.2000	0.7765
hsa_steady_inv_markup_end.2019_baseline.20260622.132122	hsa_steady	delta=0	0.1000	0.2000	104.9638
hsa_steady_inv_markup_exclude.covid_baseline.20260622.132118	hsa_steady	kappa_0=0	0.1000	0.2000	1.0619
hsa_steady_inv_markup_exclude.covid_baseline.20260622.132118	hsa_steady	delta=0	0.1000	0.2000	163.9798



data_spec	prior	parameter	mean	median	ci_2.5	ci_97.5	p_gt_0	p_lt_0	conclusion
inv_markup	baseline	alpha	0.6484	0.6341	0.5539	0.7759	1.0000	0.0000	NaN
inv_markup	baseline	kappa_0	0.1008	0.0783	-0.1878	0.4623	0.6750	0.3250	NaN
inv_markup	baseline	delta	-0.0006	-0.0007	-0.0028	0.0021	0.3000	0.7000	NaN
inv_markup	baseline	rho_1	1.8054	1.7946	1.7062	1.8953	1.0000	0.0000	NaN
inv_markup	baseline	rho_2	-0.8090	-0.7989	-0.8973	-0.7095	0.0000	1.0000	NaN
inv_markup	baseline	sigma_e	0.6785	0.6781	0.6219	0.7502	1.0000	0.0000	NaN
inv_markup	baseline	sigma_zeta	0.5045	0.4964	0.4495	0.5570	1.0000	0.0000	NaN
inv_markup	baseline	sigma_u	0.5105	0.4864	0.4296	0.6400	1.0000	0.0000	NaN
inv_markup	baseline	sigma_eps	0.3901	0.3884	0.3255	0.5235	1.0000	0.0000	NaN
inv_markup	baseline	sigma_N	0.2989	0.2976	0.2604	0.3341	1.0000	0.0000	NaN
inv_markup	weak	alpha	0.6458	0.6486	0.5432	0.7534	1.0000	0.0000	NaN
inv_markup	weak	kappa_0	0.2063	0.1583	-0.7736	1.3392	0.6250	0.3750	NaN
inv_markup	weak	delta	-0.0009	-0.0009	-0.0066	0.0041	0.3750	0.6250	NaN
inv_markup	weak	rho_1	1.9069	1.9105	1.8328	1.9713	1.0000	0.0000	NaN
inv_markup	weak	rho_2	-0.9096	-0.9116	-0.9730	-0.8367	0.0000	1.0000	NaN
inv_markup	weak	sigma_e	0.6771	0.6717	0.6029	0.7878	1.0000	0.0000	NaN
inv_markup	weak	sigma_zeta	0.5022	0.4988	0.4480	0.5509	1.0000	0.0000	NaN
inv_markup	weak	sigma_u	0.4183	0.4162	0.3521	0.4827	1.0000	0.0000	NaN
inv_markup	weak	sigma_eps	0.3040	0.3035	0.2409	0.3625	1.0000	0.0000	NaN
inv_markup	weak	sigma_N	0.2268	0.2215	0.1957	0.2747	1.0000	0.0000	NaN
inv_markup	tight	alpha	0.6124	0.6078	0.5147	0.7208	1.0000	0.0000	NaN
inv_markup	tight	kappa_0	0.0996	0.1066	-0.1344	0.2868	0.8750	0.1250	NaN
inv_markup	tight	delta	-0.0008	-0.0007	-0.0023	0.0005	0.1750	0.8250	NaN
inv_markup	tight	rho_1	0.5521	0.5496	0.3924	0.6863	1.0000	0.0000	NaN
inv_markup	tight	rho_2	-0.3597	-0.3573	-0.5528	-0.1244	0.0000	1.0000	NaN
inv_markup	tight	sigma_e	0.6758	0.6708	0.6194	0.7495	1.0000	0.0000	NaN
inv_markup	tight	sigma_zeta	0.4880	0.4869	0.4428	0.5439	1.0000	0.0000	NaN
inv_markup	tight	sigma_u	0.5543	0.5562	0.4690	0.6687	1.0000	0.0000	NaN
inv_markup	tight	sigma_eps	1.2632	1.2487	1.1261	1.4178	1.0000	0.0000	NaN
inv_markup	tight	sigma_N	0.5328	0.5308	0.4439	0.6412	1.0000	0.0000	NaN
inv_markup	all	delta_robustness	NaN	NaN	NaN	NaN	NaN	NaN	robust
output_gap_bn	baseline	alpha	0.6552	0.6674	0.5340	0.7545	1.0000	0.0000	NaN
output_gap_bn	baseline	kappa_0	0.1364	0.1811	-0.2435	0.4295	0.8000	0.2000	NaN
output_gap_bn	baseline	delta	-0.0004	-0.0006	-0.0017	0.0016	0.2500	0.7500	NaN
output_gap_bn	baseline	rho_1	1.8025	1.7995	1.7002	1.8931	1.0000	0.0000	NaN
output_gap_bn	baseline	rho_2	-0.8061	-0.8033	-0.8996	-0.7028	0.0000	1.0000	NaN
output_gap_bn	baseline	sigma_e	0.6683	0.6673	0.6126	0.7439	1.0000	0.0000	NaN
output_gap_bn	baseline	sigma_zeta	0.5837	0.5794	0.5209	0.6448	1.0000	0.0000	NaN
output_gap_bn	baseline	sigma_u	0.5135	0.5022	0.4415	0.6337	1.0000	0.0000	NaN
output_gap_bn	baseline	sigma_eps	0.3848	0.3797	0.3123	0.4833	1.0000	0.0000	NaN
output_gap_bn	baseline	sigma_N	0.2996	0.3019	0.2596	0.3365	1.0000	0.0000	NaN
output_gap_bn	weak	alpha	0.6744	0.6902	0.5567	0.7893	1.0000	0.0000	NaN
output_gap_bn	weak	kappa_0	0.3115	0.3900	-0.3883	1.0701	0.7500	0.2500	NaN
output_gap_bn	weak	delta	-0.0013	-0.0016	-0.0054	0.0028	0.3000	0.7000	NaN
output_gap_bn	weak	rho_1	1.9131	1.9078	1.8481	1.9768	1.0000	0.0000	NaN
output_gap_bn	weak	rho_2	-0.9172	-0.9136	-0.9796	-0.8540	0.0000	1.0000	NaN
output_gap_bn	weak	sigma_e	0.6544	0.6444	0.5987	0.7424	1.0000	0.0000	NaN
output_gap_bn	weak	sigma_zeta	0.5774	0.5756	0.5283	0.6438	1.0000	0.0000	NaN
output_gap_bn	weak	sigma_u	0.4021	0.4096	0.3227	0.4756	1.0000	0.0000	NaN
output_gap_bn	weak	sigma_eps	0.2940	0.2877	0.2353	0.3980	1.0000	0.0000	NaN
output_gap_bn	weak	sigma_N	0.2335	0.2293	0.2019	0.2807	1.0000	0.0000	NaN
output_gap_bn	tight	alpha	0.6354	0.6330	0.5449	0.7738	1.0000	0.0000	NaN
output_gap_bn	tight	kappa_0	0.1386	0.1442	-0.0929	0.3230	0.8750	0.1250	NaN
output_gap_bn	tight	delta	-0.0005	-0.0005	-0.0020	0.0009	0.1750	0.8250	NaN
output_gap_bn	tight	rho_1	0.5635	0.5612	0.3515	0.7331	1.0000	0.0000	NaN
output_gap_bn	tight	rho_2	-0.3716	-0.3916	-0.5634	-0.1618	0.0000	1.0000	NaN
output_gap_bn	tight	sigma_e	0.6686	0.6644	0.6191	0.7473	1.0000	0.0000	NaN
output_gap_bn	tight	sigma_zeta	0.5664	0.5628	0.5151	0.6292	1.0000	0.0000	NaN
output_gap_bn	tight	sigma_u	0.5522	0.5539	0.4803	0.6443	1.0000	0.0000	NaN
output_gap_bn	tight	sigma_eps	1.2563	1.2557	1.1292	1.4280	1.0000	0.0000	NaN
output_gap_bn	tight	sigma_N	0.5393	0.5354	0.4411	0.6388	1.0000	0.0000	NaN
output_gap_bn	all	delta_robustness	NaN	NaN	NaN	NaN	NaN	NaN	robust
unemployment_gap	baseline	alpha	0.6577	0.6608	0.5494	0.7578	1.0000	0.0000	NaN
unemployment_gap	baseline	kappa_0	0.0757	0.0893	-0.3673	0.3665	0.7000	0.3000	NaN
unemployment_gap	baseline	delta	-0.0001	-0.0001	-0.0017	0.0021	0.4750	0.5250	NaN
unemployment_gap	baseline	rho_1	1.8134	1.8142	1.7138	1.9009	1.0000	0.0000	NaN
unemployment_gap	baseline	rho_2	-0.8168	-0.8150	-0.9047	-0.7165	0.0000	1.0000	NaN
unemployment_gap	baseline	sigma_e	0.6668	0.6589	0.6003	0.7330	1.0000	0.0000	NaN
unemployment_gap	baseline	sigma_zeta	0.5877	0.5907	0.5293	0.6718	1.0000	0.0000	NaN
unemployment_gap	baseline	sigma_u	0.5017	0.5123	0.4005	0.5937	1.0000	0.0000	NaN
unemployment_gap	baseline	sigma_eps	0.3803	0.3806	0.3130	0.4622	1.0000	0.0000	NaN
unemployment_gap	baseline	sigma_N	0.3044	0.3028	0.2567	0.3591	1.0000	0.0000	NaN
unemployment_gap	weak	alpha	0.6432	0.6343	0.5567	0.7306	1.0000	0.0000	NaN
unemployment_gap	weak	kappa_0	-0.6887	-0.7053	-1.1713	-0.1801	0.0000	1.0000	NaN
unemployment_gap	weak	delta	0.0041	0.0045	0.0015	0.0066	1.0000	0.0000	NaN

note

Run scripts/05_period_robustness.py first.

11 Posterior Predictive Checks and Figures

Posterior predictive checks are generated after model runs.

12 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the sampler units and supplied data.